Chapter 8, Lecture 13

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Sufficiency

Throughout, let $X_1, \ldots, X_n$ be iid $f(x|\theta)$.

Theorem Let T be a sufficient statistic of θ. Then the MLE is a function of T, that is,

$$\widehat{\theta}_{MLE} = \widehat{\theta}_{MLE}(T).$$

This is a direct consequence of the factorization theorem, as the likelihood $\text{lik}(\theta)$ factorizes into $h(\mathbf{x})$ and $g(T(\mathbf{x}), \theta)$. As a result,

$$\max_\theta \text{lik}(\theta) = \max_\theta h(\mathbf{x}) g(T(\mathbf{x}), \theta) = h(\mathbf{x}) \max_\theta g(T(\mathbf{x}), \theta)$$

and we see that the maximizer of the likelihood function is indeed a function of T.
We have seen this several times. For instance. if we sample from a Exponential distribution with unknown parameter λ, the MLE is a function of the sufficient statistic $T = \sum_{i=1}^n X_i$, as $\widehat{\lambda} = n/T$.
An important result is the following theorem, due to Rao-Blackwell (page 310 in your textbook). In words, it says that if an estimate is *not* based on a sufficient statistic, it can be improved. It also tells you how to improve on it – by conditioning on the sufficient statistic.

Intermezzo. We need a few reminders of conditional expectations and variances. See page 147. First, $g(T) = \mathbb{E}[X|T]$ is a random variable with mean

$$\mathbb{E}[\mathbb{E}[X|T]] = \mathbb{E}[X]$$

For the discrete case, this follows from

$$\begin{aligned}
\mathbb{E}[[\mathbb{E}[X|T]] &= \mathbb{E}[g(T)] \\
&= \sum_t g(t)\mathbb{P}\{T=t\} \\
&= \sum_t \left(\sum_i x_i \mathbb{P}(X=x_i|T=t)\right)\mathbb{P}(T=t) \\
&= \sum_i x_i \mathbb{P}(X=x_i) \\
&= \mathbb{E}[X]
\end{aligned}$$

In addition, we have the formula,

$$\begin{aligned}
\mathrm{var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \\
&= \mathbb{E}[\mathbb{E}[X^2|T]] - (\mathbb{E}[\mathbb{E}[X|T]])^2 \\
&= \left\{\mathbb{E}[\mathbb{E}[X^2|T]] - \mathbb{E}[(\mathbb{E}[X|T])^2]\right\} + \left\{\mathbb{E}[(\mathbb{E}[X|T])^2] - (\mathbb{E}[\mathbb{E}[X|T]])^2\right\} \\
&:= \mathbb{E}[\mathrm{Var}(X|T)] + \mathrm{Var}(\mathbb{E}[X|T])
\end{aligned}$$

Theorem *Let $\widehat{\theta}$ be an estimator of θ with $\mathbb{E}[\widehat{\theta}^2] < \infty$. Let T be a sufficient statistic of θ. Define*

$$\widetilde{\theta} := \mathbb{E}[\widehat{\theta}|T].$$

Then, we have

(a) *$\widetilde{\theta}$ does not depend on θ*

(b) *MSE($\widetilde{\theta}$) $\leq$MSE($\widehat{\theta}$).*

Proof: The conditional distribution of $\mathbf{X}$ given $T = t$ doesn't depend on θ for every t because T is sufficient. This implies that $\widehat{\theta} = \widehat{\theta}(\mathbf{X})$ given $T = t$ doesn't depend on θ, and in turn, that the conditional expectation of $\widehat{\theta}$ given T is independent of θ. That is, (a) follows, and we can view $\widetilde{\theta}$ as a bona-fide estimator.

For part (b), we note that by the law of total expectation, we have

$$\mathbb{E}[\widetilde{\theta}] = \mathbb{E}[\mathbb{E}[\widehat{\theta}|T]] = \mathbb{E}[\widehat{\theta}]$$

meaning that $\widehat{\theta}$ and $\widetilde{\theta}$ have the same bias. In order to compare their MSEs, it remains to compare their variances. The formula we derived earlier comes in handy, as

$$\begin{aligned} \mathrm{Var}(\widehat{\theta}) &= \mathrm{Var}(\mathbb{E}[\widehat{\theta}|T]) + \mathbb{E}[\mathrm{Var}(\widehat{\theta}|T)] \\ &= \mathrm{Var}(\widetilde{\theta}) + \mathbb{E}[\mathrm{Var}(\widehat{\theta}|T)] \\ &\geq \mathrm{Var}(\widetilde{\theta}). \end{aligned}$$

This proves claim (b). ■

Minimal Sufficiency*

We have seen that a sufficient statistic is not unique – any 1-1 function of a sufficient statistic is again sufficient. What is more troubling though, is that $\mathbf{X} = (X_1, \ldots, X_n)$ itself is sufficient. There is no reduction whatsoever.... To address this issue, we introduce the concept of *minimal* sufficiency. We try to find the statistic that most parsimoniously captures all the information about θ.

Definition A sufficient statistic T is called *minimal sufficient* iff it is a function of any other sufficient statistic.

We give a constructive characterization of a minimal sufficient statistic as follows. First we define an equivalence class of sample points $\mathbf{x} \in \mathbb{R}^n$.

$$\mathbf{x} \sim \mathbf{y} : \Longleftrightarrow \ \frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} \ \text{ does not depend on } \theta$$

Define T such that

$$T(\mathbf{x}) = T(\mathbf{y}) \iff \mathbf{x} \sim \mathbf{y}$$
$$T(\mathbf{x}) \neq T(\mathbf{y}) \iff \mathbf{x} \not\sim \mathbf{y}$$

Claim: This T is minimal sufficient.
Proof: We cover only the discrete case. T is sufficient since

$$\mathbb{P}\{\mathbf{X} = \mathbf{x} | T = t\} = 0$$

if $T(\mathbf{x}) \neq t$. Otherwise,

$$\begin{aligned}
\mathbb{P}\{\mathbf{X} = \mathbf{x} | T = t\} &= \frac{\mathbb{P}\{\mathbf{X} = \mathbf{x}\}}{\sum_{\mathbf{y}:\ T(\mathbf{y})=t} \mathbb{P}\{\mathbf{X} = \mathbf{y}\}} \\
&= \frac{f(\mathbf{x}|\theta)}{\sum_{\mathbf{y}:\ T(\mathbf{y})=t} f(\mathbf{y}|\theta)} \\
&= \left(\sum_{\mathbf{y}:\ T(\mathbf{y})=t} \frac{f(\mathbf{y}|\theta)}{f(\mathbf{x}|\theta)} \right)^{-1}
\end{aligned}$$

which is independent of θ, as $T(\mathbf{x}) = t = T(\mathbf{y})$ means $\mathbf{x} \sim \mathbf{y}$. In order to show minimal sufficiency, let S be another sufficient statistic. Suppose $\mathbf{x}$ and $\mathbf{y}$ are such that $S(\mathbf{x}) = S(\mathbf{y})$. By the factorization theorem and the fact that S is sufficient, we have

$$\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \frac{g(S(\mathbf{x}), \theta)h(\mathbf{x})}{g(S(\mathbf{y}), \theta)h(\mathbf{y})} = \frac{h(\mathbf{x})}{h(\mathbf{y})}$$

which is independent of θ. This implies that $T(\mathbf{x}) = T(\mathbf{y})$, and consequently T is a function of S. ■

EXAMPLE Let $X_1, \ldots, X_n$ be independent binary trials with probability of success p. From the calculation

$$\frac{f(\mathbf{x}|p)}{f(\mathbf{y}|p)} = p^{\sum_i x_i - \sum_i y_i} (1-p)^{\sum_i y_i - \sum_i x_i}$$

we see that $f(\mathbf{x}|p)/f(\mathbf{y}|p)$ does not depend on p if and only if $\sim_i x_i = \sum_i y_i$. Conclusion: $T = \sum_{i=1}^n X_i$ is a minimal sufficient statistic for p.

EXAMPLE Let $X_1, \ldots, X_n$ be independent $N(\mu, \sigma^2)$. Show that $(\overline{X}_n, S^2)$ is minimal sufficient.

Use

$$f(\mathbf{x}|\mu, \sigma^2) = (2\pi)^{-n/2}\sigma^{-n} \exp\left(-\sum_{i=1}^n (x_i - \mu)^2/(2\sigma^2\right)$$

and the identity

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n (x_i - \overline{x}_n)^2 + n(\overline{x}_n - \mu)^2.$$

to deduce that

$$\frac{f(\mathbf{x}|\mu, \sigma^2)}{f(\mathbf{y}|\mu, \sigma^2)} = \exp\left(-\frac{1}{2\sigma^2}(n(\overline{x}_n - \mu)^2 - n(\overline{y}_n - \mu)^2)\right) \times \exp\left(-\frac{1}{2\sigma^2}(\sum_{i=1}^n (x_i - \overline{x}_n)^2 - \sum_{i=1}^n (y_i - \overline{y}_n)^2)\right)$$

The RHS does not depend on the pair (μ, σ^2) if and only if $\overline{x}_n = \overline{y}_n$ and $S_x^2 = S_y^2$.

We forego other concepts such as completeness, which are sufficient (but not necessary) for minimal sufficiency, and allow for finding UMVUEs.